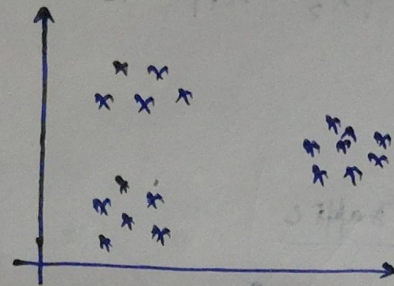
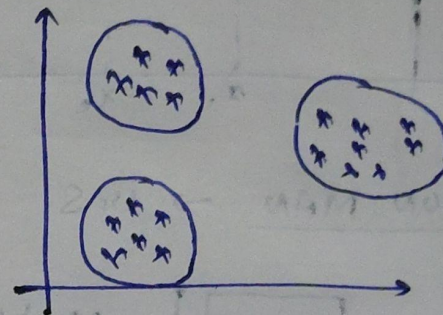


* Hierarchical clustering.

M. 89



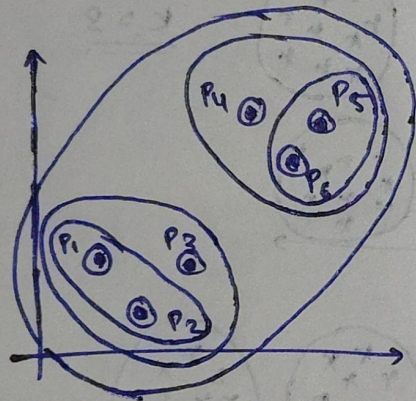
→ Hierarchical clustering



$K=3$ No. of centroids.

Hierarchical clustering

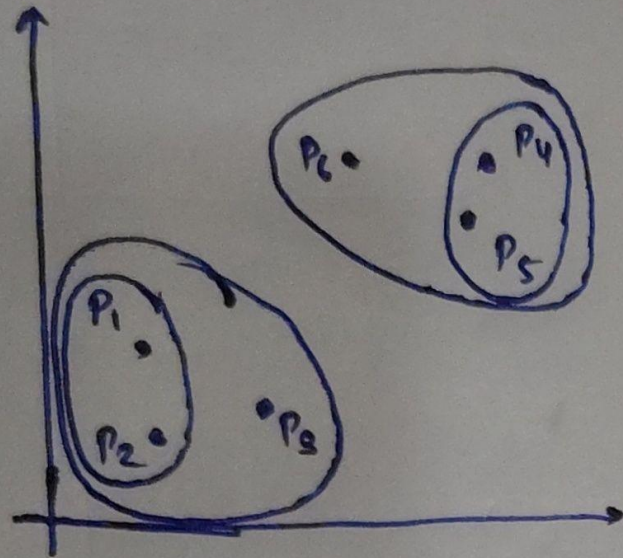
- i. Agglomerative
 - ii. Divisive
- } ⇒ Geometric intuition



Steps:

- i. For each point initially will consider it as a separate cluster.
- ii. Find the nearest point and create a new cluster.
- iii. Keep on doing the same process until we get a single cluster.

Dendrogram

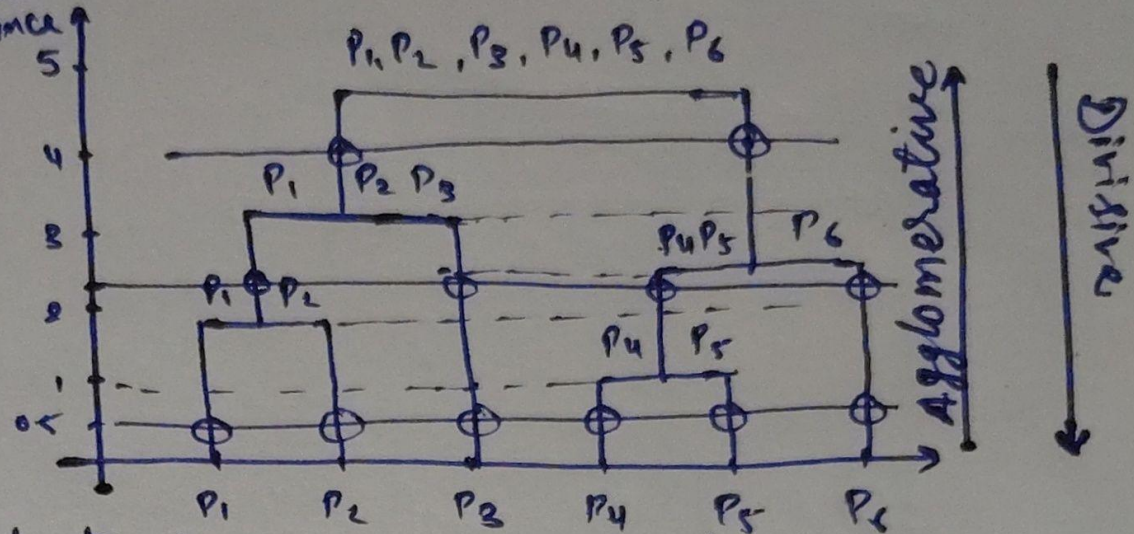


No of clusters??

Threshold
 $(K > 2)$

Dendrogram

Euclidian distance



Select the longest vertical line such that no horizontal line passes through it.
Threshold & Euclidean Distance.